

Exact asymptotic moduli in a Daugavet subspace of L^1

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September 27, 2026

Abstract

For an infinite-dimensional real subspace of L^1 whose unit ball is totally bounded in measure and whose norm has the Daugavet property, we compute the averaged midpoint and one-sided asymptotic moduli at every unit center. They are $\max\{t/2, t-1\}$ and $\max\{0, t-2\}$, respectively. The same weak-neighborhood geometry excludes every equivalent asymptotically uniformly convex norm. A quantitative realization of the Kadets–Werner construction supplies a space with both hypotheses.

1 Introduction

How much can two-sided midpoint growth differ from one-sided asymptotic convexity? We answer this quantitatively for certain real subspaces of L^1 . The norm is fixed throughout the computation: it is the inherited L^1 norm. The obstruction to AUC renorming will require a separate argument that applies to every equivalent norm.

For an infinite-dimensional real Banach space X , write B_X and S_X for its closed unit ball and unit sphere, and $\text{cof}(X)$ for its closed linear subspaces of finite codimension. Given a norm N , a vector x with $N(x) = 1$, and $t > 0$, set

$$H_N(x, t) = \sup_{F \in \text{cof}(X)} \inf_{\substack{y \in F \\ N(y) \geq 1}} \left(\frac{N(x + ty) + N(x - ty)}{2} - 1 \right),$$
$$D_N(x, t) = \sup_{F \in \text{cof}(X)} \inf_{\substack{y \in F \\ N(y) = 1}} (N(x + ty) - 1).$$

The averaged midpoint modulus $\widehat{\delta}_N(t)$ and the one-sided modulus $\bar{\delta}_N(t)$ are their respective infima over unit centers. Thus a subspace of finite codimension is chosen for each center before the perturbing direction is chosen. The norm N is *asymptotically uniformly convex* (AUC) if $\bar{\delta}_N(t) > 0$ for every $t > 0$.

For each fixed y , both the average and the maximum of $N(x + ry)$ and $N(x - ry)$ are even convex functions of r . Thus unit directions give the same inner infimum in H_N and in its maximum-based counterpart. Let m_N be the modulus obtained by replacing the average in H_N by the maximum and then taking the infimum over unit centers. It satisfies $\frac{1}{2}m_N(t) \leq \widehat{\delta}_N(t) \leq m_N(t)$. Indeed, the upper bound is immediate; for the lower bound, intersect a candidate subspace with the kernel of a norm-one functional norming x . Both perturbed norms are then at least one, and the average of their excesses is at least half their maximum excess. Taking the infima and suprema gives the comparison. Positivity of the averaged modulus is therefore the usual asymptotic midpoint uniform convexity (AMUC).

Dilworth, Kutzarova, Randrianarivony, Revalski, and Zhivkov introduced AMUC and asked whether AMUC and AUC are equivalent up to renorming [5, Definition 2.2 and the question in Section 5]. Baudier answered this question negatively by proving AMUC for subspaces of L^1 whose unit balls are relatively compact in measure and applying this to the Daugavet example of Kadets and Werner [2, Theorem 1 and Corollary 1]. We calculate both moduli exactly under these two geometric hypotheses. Baudier's proof uses the L^1 splitting identity of Brézis and Lieb [4, Theorem 1] along almost-everywhere convergent subsequences. Our lower estimate instead uses a finite net of pointwise truncations and the sign functionals attached to that net.

Recall that a Banach space has the *Daugavet property* if $\|I + T\| = 1 + \|T\|$ for every rank-one bounded operator T on the space. For functions on a probability space, convergence in measure is induced by the metric

$$d(f, g) = \inf\{a > 0 : \mathbb{P}(|f - g| > a) < a\}. \quad (1)$$

Total boundedness for this metric means that every positive radius admits a finite covering by balls of that radius. The following theorem combines this compactness assumption with the Daugavet property.

Theorem 1.1. *Let E be an infinite-dimensional closed real subspace of $L^1(\Omega, \mathbb{P})$, where $\mathbb{P}$ is a probability measure. Suppose its closed unit ball is totally bounded for convergence in measure and its inherited norm has the Daugavet property. Then every $x \in S_E$ and $t > 0$ satisfy*

$$H_E(x, t) = \max\{t/2, t - 1\}, \quad D_E(x, t) = \max\{0, t - 2\}.$$

Consequently these are also the formulas for $\widehat{\delta}_E(t)$ and $\bar{\delta}_E(t)$. The space E admits no equivalent AUC norm.

In particular, the averaged modulus has linear growth near the origin, whereas the one-sided modulus remains zero up to $t = 2$. Baudier also gives a universal linear lower bound for the maximum-based modulus of the Kadets–Werner space [2, final paragraph]. Our refinement determines the exact averaged and one-sided curves at every unit center under the two geometric hypotheses. Figure 1 displays the two curves.

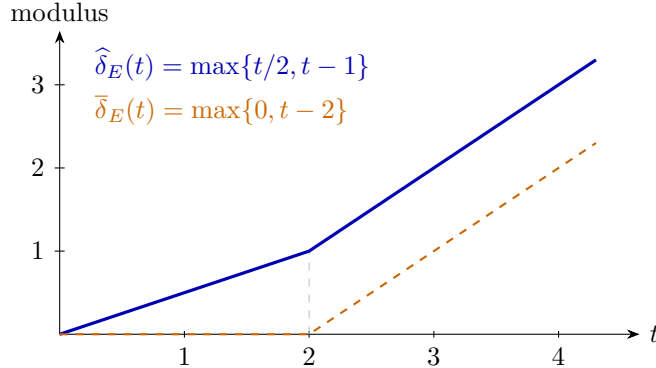


Figure 1: The exact moduli of the original norm on E . The averaged midpoint modulus is positive for every positive radius, whereas the one-sided modulus vanishes through radius 2. Theorem 1.1 gives both equalities at each unit center. These curves do not describe other equivalent norms.

The existence of such spaces has an older history. Bourgain and Rosenthal constructed an L^1 subspace whose unit ball is relatively compact in measure although the space fails the Radon–Nikodym property [3]. Kadets and Werner refined its finite-dimensional enlargement to obtain

the Daugavet property [6, Lemma 2.4 and Theorem 2.5]. We give a complete realization of their enlargement and limiting scheme in Section 6, so the existence of an example and the calculation of its moduli form one self-contained argument.

The two assumptions enter different parts of the proof. In Section 2, pointwise clipping at height $|x|/t$ turns the measure-precompact ball into a norm-compact set. Finitely many sign functionals then force at least half the L^1 mass of every unit direction in their common kernel to remain outside the clip, up to an arbitrarily small error. This gives the lower averaged bound. Section 3 develops the other input: every weak neighborhood of a unit vector contains points almost at distance two from it. This is the needed case of Shvydkoy's weak-open characterization of the Daugavet property [8, Lemma 3], obtained from the slice characterization of Kadets, Shvidkoy, Sirotkin, and Werner [7, Lemma 2.2]. In Section 4, correcting a finite-dimensional quotient coordinate turns those almost diametral differences into directions in any prescribed subspace of finite codimension, giving both upper curves. Section 5 uses the same weak-neighborhood property at a near-maximizer of an arbitrary equivalent norm to rule out AUC renorming. Section 6 then constructs an increasing sequence of finite-dimensional spaces whose closed union has both required properties. At each step, multiplying selected old unit vectors by independent positive variables creates new vectors almost at distance two from every old unit vector, while averages of the new vectors recover the selected ones. A uniform estimate places the entire enlarged unit ball close in measure to the old ball. Summable errors preserve total boundedness through the limit, and the averaging property gives the Daugavet slice condition.

2 Compact clips leave half the mass outside

In this section, E is an infinite-dimensional closed real subspace of a probability L^1 space, with its inherited norm. The clipping argument uses only total boundedness of B_E in measure.

For $x \in S_E$, $y \in E$, and $t > 0$, the real scalar identity $(|a + b| + |a - b|)/2 = \max\{|a|, |b|\}$ gives

$$\frac{\|x + ty\|_1 + \|x - ty\|_1}{2} - 1 = \int (t|y| - |x|)_+ d\mathbb{P} = t\|y\|_1 - \int \min\{|x|, t|y|\} d\mathbb{P}. \quad (2)$$

Here $r_+ = \max\{r, 0\}$. The lower bound therefore amounts to controlling the overlap with x .

Lemma 2.1. *Let E be an infinite-dimensional closed real subspace of $L^1(\Omega, \mathbb{P})$, where $\mathbb{P}$ is a probability measure. If its unit ball B_E is totally bounded in measure, then $H_E(x, t) \geq \max\{t/2, t-1\}$ for every $x \in S_E$ and $t > 0$.*

Proof. Fix x and t . Clip a function y at the pointwise height $|x|/t$:

$$C(y) = \operatorname{sgn}(y) \min\{|y|, |x|/t\}.$$

The clipped functions need not belong to E . Their norm closure $K = \overline{C(B_E)}^{L^1}$ is compact. Indeed, every sequence in B_E has a subsequence Cauchy in measure. Clipping is pointwise 1-Lipschitz, so the clipped subsequence is also Cauchy in measure. For two of its terms b_i, b_j and any $a > 0$,

$$\|b_i - b_j\|_1 \leq a + \int_{\{|b_i - b_j| > a\}} 2|x|/t d\mathbb{P}.$$

Choose a small, then use absolute continuity of the integral and the measure-Cauchy property. This proves that the subsequence is Cauchy in L^1 . Completeness proves the compactness assertion.

Fix $\varepsilon > 0$ and take a finite ε -net $b_1, \dots, b_m$ of K . Set $\sigma_i = \text{sgn}(b_i)$ and

$$F = \bigcap_{i=1}^m \ker \left(y \mapsto \int \sigma_i y \, d\mathbb{P} \right) \subseteq E.$$

This is a closed subspace of finite codimension. For $y \in S_F$, choose i with $\|C(y) - b_i\|_1 < \varepsilon$. Clipping preserves signs and reduces absolute values, so $\|y - C(y)\|_1 = 1 - \|C(y)\|_1$. Since $\int \sigma_i y = 0$, we obtain

$$\|C(y)\|_1 - \varepsilon \leq \|b_i\|_1 = \left| \int \sigma_i (b_i - y) \, d\mathbb{P} \right| \leq 1 - \|C(y)\|_1 + \varepsilon.$$

Thus $\|C(y)\|_1 \leq 1/2 + \varepsilon$. Equation (2) now gives averaged increment at least $t/2 - t\varepsilon$ on S_F . The same bound holds for $\|y\|_1 \geq 1$ by radial monotonicity. Taking the supremum over F and letting $\varepsilon \downarrow 0$ gives $H_E(x, t) \geq t/2$. The subspace is allowed to change with ε ; this step does not assert that one subspace attains the limiting bound.

For every $\|y\|_1 \geq 1$, the triangle inequality also gives

$$\frac{\|x + ty\|_1 + \|x - ty\|_1}{2} - 1 \geq t\|y\|_1 - 1 \geq t - 1.$$

Combining the two bounds proves the lemma. $\square$

Remark 2.2 (A separated-sequence proof). There is also a short contradiction argument giving the same half-mass bound. Fix a unit center x , a radius $t > 0$, and $0 < a < 1/2$. Suppose no subspace of finite codimension had averaged increment at least at in every direction of norm at least one. By radial monotonicity and Equation (2), every subspace of finite codimension would then contain a unit vector y with $\|y - C(y)\|_1 < a$. Inductively choose such y_j in the kernels of norm-one functionals norming the preceding vectors. Then $\|y_i - y_j\|_1 \geq 1$ for $i \neq j$. Compactness of the clipped set gives a subsequence whose clips are Cauchy in norm, whereas along this subsequence

$$\|y_i - y_j\|_1 \leq \|y_i - C(y_i)\|_1 + \|C(y_i) - C(y_j)\|_1 + \|C(y_j) - y_j\|_1 < 2a + o(1) < 1,$$

a contradiction for sufficiently late pairs. Thus this argument yields $H_E(x, t) \geq t/2$ by taking the supremum over the corresponding subspaces and then letting $a \uparrow 1/2$. The finite-net proof above identifies an explicit finite family of functionals that removes the overlap.

3 Almost diametral points in weak neighborhoods

To find almost diametral points subject to finitely many linear constraints, we pass from slices to relatively weakly open subsets of the ball. These points will supply both the upper bounds for the moduli and the obstruction to AUC renorming. A *slice* of B_X is a nonempty set

$$H = \{y \in B_X : \ell(y) > \alpha\}, \quad \ell \in X^*, \quad \alpha \in \mathbb{R}.$$

We first prove the geometric characterization of the Daugavet property due to Kadets, Shvidkoy, Sirotkin, and Werner [7, Lemma 2.2], also recalled by Kadets and Werner [6, Lemma 1.1].

Lemma 3.1 (Daugavet slice characterization). *A nonzero real Banach space X has the Daugavet property if and only if, for every $w \in S_X$, every $\rho > 0$, and every slice H of B_X , there is $y \in H$ with $\|w + y\| \geq 2 - \rho$.*

Proof. Suppose first that X has the Daugavet property. If the functional defining H is zero, then $H = B_X$ and we may take $y = w$. Otherwise, after dividing the functional and threshold by its norm, write $H = \{y \in B_X : \ell(y) > \alpha\}$ with $\|\ell\| = 1$ and $\alpha < 1$. Take $0 < \eta < \min\{1 - \alpha, \rho/2\}$. The rank-one operator $Ty = \ell(y)w$ has norm one, so there is $y \in S_X$ with $\|y + \ell(y)w\| > 2 - \eta$. Replacing y by $-y$ if necessary makes $\ell(y) \geq 0$ without changing this inequality. It follows that $\ell(y) > 1 - \eta > \alpha$ and

$$\|y + w\| \geq \|y + \ell(y)w\| - (1 - \ell(y)) > 2 - 2\eta > 2 - \rho.$$

Conversely, assume the slice condition and let $Tz = \ell(z)v$ be a nonzero rank-one operator with $\|\ell\| = 1$. Set $a = \|v\| = \|T\|$ and $w = v/a$. For $0 < \eta < 1$, the slice condition provides $y \in B_X$ with $\ell(y) > 1 - \eta$ and $\|w + y\| \geq 2 - \eta$. A norm-one functional φ norming $w + y$ takes value at least $1 - \eta$ on each summand. Consequently,

$$\|I + T\| \geq \|y + Ty\| \geq \varphi(y) + a\ell(y)\varphi(w) \geq (1 - \eta) + a(1 - \eta)^2.$$

Let $\eta \downarrow 0$ and use $\|I + T\| \leq 1 + \|T\|$. The identity is immediate when $T = 0$. $\square$

The passage from slices to weak neighborhoods uses Bourgain's lemma on convex combinations of slices (see [1] for the attribution). We give a finite-dimensional proof. Its use in passing to weak neighborhoods follows the same pattern as Shvydkoy's proof of the weak-open characterization [8, Lemma 3].

Lemma 3.2 (Bourgain's lemma). *Every nonempty relatively weakly open subset U of a real Banach space's closed unit ball contains a convex combination of nonempty slices of that ball.*

Proof. Choose $u \in U$, a continuous linear map $T : X \rightarrow \mathbb{R}^k$, and an open set $O \subseteq \mathbb{R}^k$ such that $u \in B_X \cap T^{-1}(O) \subseteq U$. Put $D = \overline{T(B_X)}$. We want to place a convex combination of small slices of D inside O and then pull them back to B_X . For this it suffices to approximate $T(u)$ by convex combinations of exposed points, that is, unique maximizers of linear functionals on D . We first verify that D is the closed convex hull of those points. Put $M = \max_{q \in D} |q|$, using a Euclidean norm. For a unit vector a and $R > M$, choose $p_R \in D$ farthest from $-Ra$. Expanding the squares in $|q + Ra| \leq |p_R + Ra|$ gives

$$2(p_R + Ra) \cdot (q - p_R) \leq -|q - p_R|^2 \quad (q \in D).$$

Thus p_R is the unique maximizer of the linear functional with coefficient $p_R + Ra$. If q_a maximizes $a \cdot q$ on D , the same comparison gives

$$0 \leq a \cdot q_a - a \cdot p_R \leq \frac{|p_R|^2 - |q_a|^2}{2R} \leq \frac{M^2}{2R}.$$

The exposed points therefore have the same supremum as D for every linear functional. Separation proves the closed-convex-hull assertion, also when D is a singleton.

Since $T(u) \in O$, there are exposed points $p_1, \dots, p_l$ and positive weights λ_i summing to one such that $p = \sum_i \lambda_i p_i \in O$. Choose $r > 0$ with the Euclidean ball $B(p, r) \subseteq O$. If $a_i \cdot q$ exposes p_i , compactness and uniqueness of its maximum give a threshold $\beta_i < a_i \cdot p_i$ with

$$\{q \in D : a_i \cdot q > \beta_i\} \subseteq B(p_i, r).$$

The sets $H_i = \{y \in B_X : a_i \cdot T(y) > \beta_i\}$ are nonempty slices, since $T(B_X)$ is dense in D . For $y_i \in H_i$,

$$\left| T \left(\sum_i \lambda_i y_i \right) - p \right| \leq \sum_i \lambda_i |T(y_i) - p_i| < r.$$

Their convex combination consequently belongs to U . $\square$

The slice property can now be applied successively within the slices provided by Bourgain's lemma. This gives the following specialization of Shvydkoy's weak-open result [8, Lemma 3].

Lemma 3.3. *If X has the Daugavet slice property, then, for every $x \in S_X$ and every relatively weakly open neighborhood U of x in B_X ,*

$$\sup_{v \in U} \|v - x\| = 2.$$

Proof. By Lemma 3.2, choose slices H_i and positive weights λ_i such that $\sum_i \lambda_i = 1$ and $\sum_i \lambda_i H_i \subseteq U$. Fix $0 < \eta < 1$ and set $z_0 = -x$. At step i , apply the slice property at $z_{i-1}/\|z_{i-1}\|$ to choose $y_i \in H_i$ with $\|z_{i-1}/\|z_{i-1}\| + y_i\| \geq 2 - \eta$. A norm-one functional norming this sum takes value at least $1 - \eta$ on each of its two summands. Therefore, with $z_i = z_{i-1} + \lambda_i y_i$,

$$\|z_i\| \geq (1 - \eta)(\|z_{i-1}\| + \lambda_i) > 0.$$

The induction is well defined and yields

$$\|z_l\| \geq (1 - \eta)^l + \sum_{i=1}^l \lambda_i (1 - \eta)^{l-i+1} \geq 2(1 - \eta)^l.$$

Now $v = \sum_i \lambda_i y_i \in U$ and $z_l = v - x$. Let $\eta \downarrow 0$; the reverse bound follows from $v, x \in B_X$. $\square$

4 One family of directions gives both curves

Proof of the formulas in Theorem 1.1. Lemma 2.1 gives the averaged lower bound. For the one-sided bound, choose a norm-one functional φ with $\varphi(x) = 1$. Every $y \in S_{\ker \varphi}$ satisfies

$$\|x + ty\| \geq 1, \quad \|x + ty\| \geq t - 1.$$

Hence $D_E(x, t) \geq \max\{0, t - 2\}$.

For the upper bounds, fix any $F \in \text{cof}(E)$ and choose a bounded finite-rank projection P with kernel F . Such a projection is obtained by composing the quotient map with a linear section of E/F . For $0 < \varepsilon < 1/2$,

$$U_\varepsilon = \{v \in B_E : \|P(v - x)\| < \varepsilon\}$$

is a relatively weakly open neighborhood of x . The finite-dimensional range of P makes it weak-to-norm continuous. By Lemma 3.3, choose $v_\varepsilon \in U_\varepsilon$ with $\|v_\varepsilon - x\| > 2 - \varepsilon$, and put

$$z_\varepsilon = (I - P)(v_\varepsilon - x), \quad L_\varepsilon = \|z_\varepsilon\|, \quad y_\varepsilon = z_\varepsilon/L_\varepsilon \in S_F.$$

Then

$$2 - 2\varepsilon < L_\varepsilon \leq 2 + \varepsilon, \quad \|x + z_\varepsilon\| \leq 1 + \varepsilon.$$

Thus the directions lie exactly in F , their unnormalized lengths tend to two, and the translated endpoints remain close to B_E .

For any $t > 0$, set $\lambda = \min\{t/L_\varepsilon, 1\}$. The identity

$$x + ty_\varepsilon = (1 - \lambda)x + \lambda(x + z_\varepsilon) + (t - L_\varepsilon)_+ y_\varepsilon$$

and the triangle inequality give the single bound

$$\|x + ty_\varepsilon\| \leq 1 + \varepsilon + (t - L_\varepsilon)_+.$$

The right-hand side converges to $1 + (t - 2)_+ = \max\{1, t - 1\}$. Since $\|x - ty_\varepsilon\| \leq 1 + t$, the one-sided and averaged increments have upper limits at most

$$(t - 2)_+, \quad \frac{t + (t - 2)_+}{2},$$

respectively. These are $\max\{0, t - 2\}$ and $\max\{t/2, t - 1\}$. In particular the argument includes $t = 2$, where the two branches meet. Each inner infimum for the fixed F is at most the corresponding limit. Taking the supremum over F proves the upper bounds. Both formulas hold for every unit center, so taking the infimum over centers leaves them unchanged. $\square$

5 Why a different norm cannot be AUC

The one-sided modulus vanishes for $0 < t \leq 2$, so the inherited norm is not AUC. Ruling out all equivalent norms requires the weak-neighborhood property itself. The following argument applies to any infinite-dimensional real Banach space in which every relatively weakly open neighborhood of a unit vector contains points of the unit ball arbitrarily close to distance two from that vector, as in Lemma 3.3.

Proof of the renorming assertion in Theorem 1.1. Suppose N is an equivalent AUC norm. Choose $a > 0$ with $N(u) \geq a\|u\|$ for all u , and put $R = \sup_{u \in B_E} N(u) < \infty$. Fix, in this order,

$$t_0 = \frac{a}{4R}, \quad 0 < \gamma < \bar{\delta}_N(t_0), \quad x_0 \in S_E \text{ with } (1 + \gamma)N(x_0) > R.$$

The last choice is possible since $R = \sup_{u \in S_E} N(u)$. The definition of $\bar{\delta}_N(t_0)$ at $x_0/N(x_0)$ supplies $F \in \text{cof}(E)$ such that

$$N(x_0 + z) \geq (1 + \gamma)N(x_0) \quad \text{if } z \in F, \quad N(z) \geq t_0 N(x_0). \quad (3)$$

To justify the entire range of radii, convexity gives, for $N(w) = 1$ and $r \geq t_0$,

$$N(x_0/N(x_0) + rw) - 1 \geq \frac{r}{t_0}(N(x_0/N(x_0) + t_0w) - 1).$$

Choose $0 < \eta < \min\{(1 + \gamma)N(x_0) - R, a/4\}$. The set

$$U = \{u \in B_E : \inf_{z \in F} N(u - x_0 - z) < \eta\}$$

is a relatively weakly open neighborhood of x_0 : the infimum is the norm in the finite-dimensional quotient E/F . Equivalent norms have the same continuous functionals, hence the same weak topology. For $u \in U$, choose $z \in F$ with $N(u - x_0 - z) < \eta$. If $N(z) \geq t_0 N(x_0)$, Equation (3) gives

$$N(u) > (1 + \gamma)N(x_0) - \eta > R,$$

contrary to $u \in B_E$. Thus

$$N(u - x_0) < t_0 N(x_0) + \eta \leq t_0 R + \eta < a/2, \quad \|u - x_0\| < 1/2.$$

This contradicts Lemma 3.3. $\square$

6 Constructing the measure-precompact Daugavet space

The preceding results apply to a space with two rather different properties: its unit ball is small for convergence in measure, while its slices contain vectors almost at distance two from any prescribed unit vector. We now construct such a space. The construction is due to Kadets and Werner [6, Lemma 2.4 and Theorem 2.5], who refine the method of Bourgain and Rosenthal [3]. We give all the estimates needed for the finite-dimensional enlargement and its iteration. The explicit multipliers below and the constants 10, 54, and 108 belong to the quantitative realization given here; the enlargement and iteration scheme is the credited construction.

The idea is to multiply an old unit vector by a positive random variable which has mean one but is small with high probability. Each product has small L^1 overlap with every old unit vector, so the norm of their sum is almost two. Averaging sufficiently many products recovers the old vector in norm. The main additional task is to control every vector in the new unit ball, which permits the iteration to preserve precompactness in measure.

Throughout this section,

$$(\Omega, \mathbb{P}) = ([0, 1]^{\mathbb{N}}, \lambda^{\mathbb{N}}), \quad L^1 = L^1(\Omega, \mathbb{P}; \mathbb{R}),$$

and $\|\cdot\|$ denotes the L^1 norm. Write $B(G)$ and $S(G)$ for the closed unit ball and unit sphere of a subspace G . A space depends on finitely many coordinates if one common finite set of coordinates determines all its elements.

Recall the metric from (1), on measurable functions modulo equality almost everywhere:

$$d(f, g) = \inf\{a > 0 : \mathbb{P}(|f - g| > a) < a\}. \quad (4)$$

This metric induces convergence in measure. For example, if $\mathbb{P}(|f - g| > a) < a$ and $\mathbb{P}(|g - h| > b) < b$, the union bound gives $\mathbb{P}(|f - h| > a + b) < a + b$; taking infima proves the triangle inequality. The identity $d(f, g) = 0$ is equivalent to $f = g$ almost everywhere, and the convergence assertion follows directly from the thresholds in (4). Markov's inequality also gives $d(f, g) \leq \|f - g\|^{1/2}$, so norm convergence implies convergence for d .

Theorem 6.1 (Kadets–Werner construction). *There is an infinite-dimensional closed real subspace $E \subseteq L^1$ such that $B(E)$ is totally bounded for d and has the following slice property: for every $w \in S(E)$, every nonempty slice*

$$H = \{y \in B(E) : \ell(y) > \alpha\}, \quad \ell \in E^*, \quad \alpha \in \mathbb{R},$$

and every $\rho > 0$, there is $y \in H \cap S(E)$ such that $\|w + y\| \geq 2 - \rho$. In particular, E has the Daugavet property.

6.1 The finite-dimensional enlargement

We need the new unit vectors to be almost additive in norm with every old unit vector, and their averages to approximate selected old vectors. The third conclusion below keeps the entire new unit ball close, in measure, to the old unit ball.

Lemma 6.2. *Suppose $G \neq \{0\}$ is a finite-dimensional subspace of L^1 depending on finitely many coordinates, $m \geq 1$, $u_1, \dots, u_m \in S(G)$, and $0 < h < 1$. There are a finite-dimensional*

$G' \supseteq G$ depending on finitely many coordinates, an integer $n \geq 1$, and vectors $v_{kj} \in S(G')$ ($1 \leq k \leq m$, $1 \leq j \leq n$) such that

$$\|w + v_{kj}\| \geq 2 - h \quad (w \in S(G), k \leq m, j \leq n), \quad (5)$$

$$\|u_k - \frac{1}{n} \sum_{j=1}^n v_{kj}\| \leq h \quad (k \leq m), \quad (6)$$

$$\text{for every } g \in B(G') \text{ there is } \psi_0 \in B(G) \text{ with } d(g, \psi_0) \leq h. \quad (7)$$

To prove the lemma, we will use independent positive multipliers whose mean is one and whose mass is concentrated near zero. The next estimate controls their linear combinations; this is what the whole-ball conclusion requires beyond control of the selected vectors themselves.

6.2 A heavy-tail estimate

The following variable has mean one for every $p > 1$, although it converges to zero in probability as $p \downarrow 1$. The estimate for differences will control arbitrary linear combinations in an enlarged unit ball.

Lemma 6.3. *Let $1 < p \leq 3/2$, $s = (p-1)/p$, and $f = sV^{-1/p}$, where V is uniform on $(0, 1)$. Let $Y_1, \dots, Y_n$ be independent variables, each distributed as the difference of two independent copies of f . For real scalars $a_1, \dots, a_n$, put $A = (\sum_{j=1}^n |a_j|^p)^{1/p}$ and $Z = \sum_{j=1}^n a_j Y_j$. Then $\mathbb{E}f = 1$ and*

$$\mathbb{P}(|Z| > u) \leq 10(sA/u)^p \quad (u > 0), \quad \mathbb{E}|Z| \geq A/54. \quad (8)$$

The lower estimate bounds A from an L^1 bound on Z ; the upper estimate then gives concentration in probability uniformly in n .

Proof. Direct integration gives $\mathbb{E}f = sp/(p-1) = 1$, and $\mathbb{P}(f > u) = (s/u)^p$ for $u \geq s$. For all $u > 0$ the same expression is an upper bound. If $Y = f - f'$ for independent copies, then Y is symmetric and $\mathbb{P}(|Y| > u) \leq 2(s/u)^p$. For fixed $u > 0$, the variables $X_j = a_j Y_j \mathbf{1}_{\{|a_j Y_j| \leq u\}}$ are independent and centered, and

$$\mathbb{E}X_j^2 \leq \int_0^u 2v \mathbb{P}(|a_j Y_j| > v) dv \leq \frac{4(s|a_j|)^p u^{2-p}}{2-p}.$$

Except on the event that some $|a_j Y_j| > u$, the sums Z and $\sum_j X_j$ agree. The union bound and Chebyshev's inequality therefore give

$$\mathbb{P}(|Z| > u) \leq \left(2 + \frac{4}{2-p}\right) (sA/u)^p \leq 10(sA/u)^p.$$

For the lower bound, when $u \geq s$, the event $\{f > u + 2s, f' \leq 2s\}$ gives

$$\mathbb{P}(|Y| > u) \geq \left(\frac{s}{u+2s}\right)^p (1 - 2^{-p}) \geq \frac{1}{18}(s/u)^p.$$

Here $u + 2s \leq 3u$, $1 - 2^{-p} \geq 1/2$, and $3^p \leq 9$. Suppose $A > 0$ and omit zero coefficients. Since $|a_j| \leq A$, independence implies, for $u \geq sA$,

$$\begin{aligned} \mathbb{P}\left(\max_j |a_j Y_j| > u\right) &\geq 1 - \exp\left(-\frac{1}{18}(sA/u)^p\right) \\ &\geq \frac{1}{36}(sA/u)^p. \end{aligned}$$

The last inequality uses $1 - e^{-v} \geq v/2$ for $0 \leq v \leq 1$. Adjoin independent signs $\varepsilon_j \in \{-1, 1\}$ of equal probability. For each fixed realization of the terms, averaging first over the sign of a term of largest absolute value gives

$$\mathbb{E}_\varepsilon \left| \sum_j \varepsilon_j a_j Y_j \right| \geq \max_j |a_j Y_j|,$$

because $(|b+c| + |b-c|)/2 \geq |c|$. The signed sum has the same law as Z , by independence and symmetry. Consequently,

$$\mathbb{E}|Z| \geq \mathbb{E} \max_j |a_j Y_j| \geq \frac{1}{36} \int_{sA}^{\infty} (sA/u)^p du = \frac{A}{36p} \geq \frac{A}{54}.$$

If $A = 0$, both assertions are immediate. □

6.3 Proof of the enlargement

The scalar estimates now let us control all linear combinations in the new space. We prove the three conclusions of Lemma 6.2 in the order that permits the number of multipliers to be chosen last.

Proof of Lemma 6.2. We first fix the distribution of the new multipliers and prove the first and third conclusions uniformly in their number. Only then will we choose the number of multipliers to obtain the averaging conclusion.

Parameter choices. Choose $c \in (0, h/4)$, and then $r, q > 0$ sufficiently small that

$$mq < 1, \quad mq + h^{-1}(2mr + (1 - mq)^{-1} - 1) < h. \quad (9)$$

With $s = (p-1)/p$, choose $1 < p \leq 3/2$ close enough to 1 that

$$s < c, \quad 2((s/c)^p + c) < h, \quad 10(108s/r)^p < q. \quad (10)$$

These choices are possible because $s \rightarrow 0$ as $p \downarrow 1$. We fix p now; the integer n will be chosen last.

Near-additivity with the old space. For any prospective n , take mn independent copies f_{kj} of the variable f in Lemma 6.3, each on a fresh coordinate, and put

$$v_{kj} = u_k f_{kj}, \quad G'_n = \text{span}(G \cup \{v_{kj} : k \leq m, j \leq n\}).$$

Independence from the old coordinates gives $\|v_{kj}\| = 1$. For every $w \in S(G)$,

$$\begin{aligned} \|w + v_{kj}\| &\geq 2 - 2\mathbb{E} \min(|w|, |u_k| f_{kj}) \\ &\geq 2 - 2(\mathbb{P}(f > c) + c) > 2 - h. \end{aligned}$$

Indeed, on $\{f_{kj} > c\}$ bound the minimum by $|w|$, and on its complement by $c|u_k|$; independence then gives the last bound. Thus (5) holds for every n .

Approximation of the whole unit ball. We next prove (7), also uniformly in n . Fix $g \in B(G'_n)$ and choose any representation

$$g = x + \sum_{k=1}^m u_k z_k, \quad x \in G, \quad z_k = \sum_{j=1}^n a_{kj} f_{kj}.$$

Fix k . On an auxiliary product probability space resample all the variables in the k th group, leaving the old coordinates and all other groups unchanged. Write z'_k for the resampled sum and $g^{(k)}$ for the resulting function. The new variables are independent copies, so $g^{(k)}$ has the same law as g , and

$$\mathbb{E}|z_k - z'_k| = \mathbb{E}|u_k(z_k - z'_k)| = \mathbb{E}|g - g^{(k)}| \leq 2.$$

The first equality uses $\|u_k\| = 1$ and independence of the old coordinates. Lemma 6.3, applied to the differences of the paired copies, therefore gives

$$A_k = \left(\sum_j |a_{kj}|^p \right)^{1/p} \leq 108, \quad \mathbb{P}(|z_k - z'_k| > r) \leq 10(108s/r)^p < q. \quad (11)$$

In particular, the bound does not depend on n or on the chosen representation of g .

The estimate for $z_k - z'_k$ says that z_k is concentrated near some constant. Indeed, averaging the conditional probability over z'_k gives a deterministic $c_k \in \mathbb{R}$ with $\mathbb{P}(|z_k - c_k| > r) < q$. These constants may depend on g ; no simultaneous choice is needed. Set

$$\psi = x + \sum_{k=1}^m c_k u_k \in G, \quad D = \bigcap_{k=1}^m \{|z_k - c_k| \leq r\}.$$

The event D depends only on the fresh coordinates, has probability at least $1 - mq$, and on D we have $|g - \psi| \leq r \sum_k |u_k|$. Its independence from the old coordinates gives

$$1 \geq \mathbb{E}(|g| \mathbf{1}_D) \geq \mathbb{P}(D)(\|\psi\| - mr), \quad \|\psi\| \leq mr + (1 - mq)^{-1}.$$

At this point ψ need not belong to the old unit ball. Set $\psi_0 = \psi / \max\{1, \|\psi\|\} \in B(G)$. The cost of this normalization is bounded by

$$\|\psi - \psi_0\| \leq mr + (1 - mq)^{-1} - 1.$$

Markov's inequality restricted to D and the bound on its complement now yield

$$\begin{aligned} \mathbb{P}(|g - \psi_0| > h) &\leq mq + h^{-1}(mr + \|\psi - \psi_0\|) \\ &\leq mq + h^{-1}(2mr + (1 - mq)^{-1} - 1) < h. \end{aligned}$$

This proves (7) for every n .

We have now proved both near-additivity and whole-ball approximation without placing an upper bound on n .

Recovering the old vectors by averages. It remains to choose n for (6). For independent copies $f_1, \dots, f_n$ of the fixed integrable variable f , write $f^M = \min\{f, M\}$. Truncating, centering, and applying the variance bound to the bounded independent copies gives

$$\mathbb{E} \left| \frac{1}{n} \sum_{j=1}^n f_j - 1 \right| \leq 2\mathbb{E}(f - f^M) + \sqrt{\text{Var}(f^M)/n}. \quad (12)$$

The first term tends to zero as $M \rightarrow \infty$; after M is fixed the second tends to zero as $n \rightarrow \infty$. Choose n so that the left side is at most h . All groups have the same law, and independence gives

$$\|u_k - \frac{1}{n} \sum_j v_{kj}\| = \mathbb{E} \left| 1 - \frac{1}{n} \sum_j f_{kj} \right| \leq h \quad (k \leq m).$$

Taking $G' = G'_n$ completes the proof. This space uses just finitely many additional coordinates; the earlier two conclusions persist because their estimates held for every finite n . $\square$

6.4 Passage to the closed space

We apply the enlargement to finer and finer nets of the old unit sphere. The summable errors keep the eventual unit ball precompact in measure; the approximation by averages ensures that every slice eventually contains one of the new unit vectors.

Proof of Theorem 6.1. Start with $G_1 = \text{span}\{1\}$ and put $h_b = 2^{-b}$. Given G_b , choose a finite h_b -net $\{u_{b,k} : 1 \leq k \leq m_b\}$ of $S(G_b)$ in norm and apply Lemma 6.2 with $h = h_b$. Denote the resulting space by G_{b+1} and its new unit vectors by $v_{b,kj}$, $1 \leq j \leq n_b$. All stages depend on finitely many coordinates, so the construction continues on Ω . Set

$$E = \overline{\bigcup_{b \geq 1} G_b}^{\|\cdot\|} \subseteq L^1.$$

Each inclusion $G_b \subseteq G_{b+1}$ is strict: if any $v_{b,kj}$ belonged to G_b , substituting $w = -v_{b,kj}$ in (5) would give $0 \geq 2 - h_b$. Thus E is infinite-dimensional.

The entire unit ball is totally bounded in measure. The union $\bigcup_b B(G_b)$ is norm dense in $B(E)$: if x_j in the algebraic union tends to $x \in B(E)$, then $x_j / \max\{1, \|x_j\|\}$ also tends to x . Repeated use of (7) gives, for $a > b$ and $g \in B(G_a)$,

$$\inf_{\psi \in B(G_b)} d(g, \psi) \leq \sum_{l=b}^{a-1} h_l \leq \sum_{l=b}^{\infty} h_l = 2^{1-b}.$$

The bound is automatic for $a \leq b$. Distance to a fixed nonempty set is a 1-Lipschitz function for d , so norm density and (4) extend it to every $g \in B(E)$. The set $B(G_b)$ is norm compact and therefore d -compact. Given $\varepsilon > 0$, choose b with $2^{1-b} < \varepsilon/2$ and a finite $\varepsilon/2$ -net of $B(G_b)$ for d . The preceding distance estimate shows that this is an ε -net of $B(E)$.

The Daugavet slice property. Now fix w, H, ℓ, α, ρ as in Theorem 6.1. There is $u \in S(E)$ with $\ell(u) > \alpha$: the supremum of a linear functional over the ball equals its supremum over the sphere, including when the functional is zero. Normalizing approximants from the increasing union gives $u^{(b)}, w^{(b)} \in S(G_b)$ with $u^{(b)} \rightarrow u$ and $w^{(b)} \rightarrow w$ in norm. Choose a net point u_{b,k_b} with $\|u_{b,k_b} - u^{(b)}\| \leq h_b$ and set

$$a_b = \frac{1}{n_b} \sum_{j=1}^{n_b} v_{b,k_b j}.$$

By (6), $\|a_b - u\| \leq 2h_b + \|u^{(b)} - u\| \rightarrow 0$. Thus $\ell(a_b) > \alpha$ for all sufficiently large b , and for each such b at least one summand $v_{b,k_b j}$ lies in H . For every summand, (5) also gives

$$\|w + v_{b,k_b j}\| \geq 2 - h_b - \|w - w^{(b)}\|.$$

Take b large enough that the last two error terms sum to less than ρ . The summand in H is the required unit vector.

Finally, Lemma 3.1 turns this slice property into $\|I + T\| = 1 + \|T\|$ for every rank-one operator T on E . $\square$

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